

Ray Sun

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Education

University of Illinois Urbana-Champaign

- M.S. in Computer Science (GPA: 4.0 / 4.0) 2019 – 2020
- B.S. in Computer Science, Minor in Mathematics (GPA: 4.0 / 4.0) 2016 – 2019

Publications

Identifiable Token Correspondence for World Models

ICML 2026

Youngin Kim*, **Ray Sun***, Inho Kim, Bumsoo Park, Hyun Oh Song

- Proposes a transformer world model for model-based reinforcement learning that exploits relationships between tokens across frames in 2D games
- Chooses between transformer outputs and previous tokens by solving an optimal transport problem

Research Experience

Seoul National University

Visiting Researcher

2024 – 2026

- Developed transformer world models for reinforcement learning

University of Illinois Urbana-Champaign

"Environmental Curriculum Learning for Efficiently Achieving Superhuman Play in Games"

Master's Thesis

2017 – 2020

- Created a new training technique for efficient reinforcement learning in StarCraft 2

Undergraduate Research Assistant

2017 – 2018

- Built an anomaly-based model to detect log manipulation
- Performed association rule mining on terabytes of server logs using pandas
- Developed hashing-based challenges in C++ with the ns-3 simulator to prevent DDoS
- Configured and managed a server cluster as a testbed using OpenStack and AWS

Work Experience

Meta

Senior Software Engineer

2023 – 2024

Software Engineer

2020 – 2023

- Built end-to-end encryption for Messenger and Instagram in C and SQL
- Increased notifications reliability by 150% by fixing memory issues on iOS
- Reduced message latency by 22% and drove incident response on the server
- Led security efforts on the client, including sanitization and hardening

Meta

Software Engineering Intern

2019

- Designed and deployed a music machine learning model using PyTorch (patent pending)

Microsoft

Software Engineering Intern

2018

- Provided Bing Ads insights by analyzing user behavior with sequential pattern mining

Software Engineering Intern

2017

- Optimized the Microsoft Whiteboard rendering engine in C#

Teaching Experience

CS 498 ITO: **Internet of Things**, Teaching Assistant

2020

CS 374: **Intro to Algorithms & Models of Computation**, Course Assistant

2019

CS 461: **Computer Security I**, Course Assistant

2019

Skills

- Languages: **Python** | **C** | **PHP** | **C#** | **Objective-C** | **SQL** | **C++** | **Java**
- Frameworks: **JAX** | **PyTorch** | **TensorFlow** | **pandas**

Key Courses

Machine Learning and Optimization

- CS 498: **Reinforcement Learning** — A
- CS 598: **Machine Learning for Signal Processing** — A+
- CS 446: **Machine Learning** — A
- CS 450: **Numerical Analysis** — A+
- MATH 484: **Nonlinear Programming** — A
- CS 440: **Artificial Intelligence** — A+
- CS 412: **Introduction to Data Mining** — A+

Theory

- CS 598: **Algorithmic Game Theory** — A
- CS 374: **Introduction to Algorithms & Models of Computation** — A+

Security

- CS 563: **Advanced Security** — CR
- CS 461: **Computer Security I** — A+
- CS 460: **Security Laboratory** — A+

Citizenship

- United States, Canada